

EOL_Instructions_for_FR5CP

FR5CP 下线检测指导规范



END OF LINE



BOSCH

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作者 Author	DU Yu	部门 Department	CC-DA/EPR5-CN
签名 Signature		日期 Date	2019.04.23

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签名 Signature		日期 Date	
签名 Signature		日期 Date	
签名 Signature		日期 Date	

释放 Release	Cai Jing	部门 Department	CC-DA/PAR-CN
签名 Signature		日期 Date	
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1 更改记录 Modifications

编号 No.	版本 Version	日期 Date	修订人 Reviser	章节 Chapter	修改描述 Description of modification
1	1.0	2019/03/02	DU Yu	all	初版 Initial version
2	1.1	2019/04/23	CUI Yan DU Yu	6.1.2	更新了角度精度 Angle accuracy updated
3	1.2	2020/03/20	ZOU Yi	6.1.1/ 6.1.2	更新流程图及操作步骤 Flowchart and operation process updated

2 参考文档 Reference Documents

No.	Version	Document
1	1.0	0265K60XXX_FR5CP_TCD_Part1_HW_Functions
2	1.0	0265K60XXX_FR5CP_TCD_Part2_HW_Characteristics.pdf
3	1.0	0265K60XXX_FR5CP_TCD_Part3_HW_Handling.doc
4	1.0	0265K60XXX_FR5CP_TCD_Part4_HW_Testing.doc
5	1.0	二维/三维图纸 2D / 3D drawings
6	1.0	客户诊断规范 Customer diagnostic specification
7	1.0	生产启动支持检查表 Launch management checklist
8	1.0	客户车辆配置参数 Customer Key information

3 缩写表 **Abbreviations**

OEM	原始设备制造商 O riginal E quipment M anufacturer
QM	质量管理 Q uality M anagement
EOL	下线检测 E nd O f L ine
DTC	诊断故障码 D iagnostic T rouble C ode
ECU	电子控制单元 E lectronic C ontrol U nit
ACC	自适应巡航控制 A daptive C ruise C ontrol
PEBS	预测性紧急制动系统 P redicative E mergency S ystems

4 概述 General

本文档用于对生产线上雷达的安装进行指导。

以下前提条件适用于整个文档

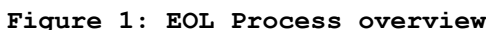
- 中距离雷达FR5CP在送往主机厂产线前已进行了预校准。
- 主机厂负责支架设计和安装精度。
- 中距离雷达FR5CP开放式安装或安装在符合规范的覆盖件后。
- 主机厂将选用静态主动式雷达校准方式。
- 主机厂需保证在覆盖件的安装不会与雷达相干涉。

This document is to be used as a guideline for the calibration of FR5CP on product line.

The following assumptions are made throughout this document:

- FR5CP is delivered preassembled and pre-aligned to the OEM plant.
- The OEM is responsible for bracket design and installation accuracy.
- The FR5CP is mounted openly or behind a validated cover.
- The OEM should follow static active alignment procedures.
- The OEM should ensure the cover installation will not conflict with FR5CP.

If an individual FR5CP calibration station is considered, then the centering machine could use vehicle physical axis as reference, but the tolerance of centering machine in combine with the tolerance of four wheel aligner must follow the alignment specification mentioned in following sub-sections.



5.1 校准的目的 Purpose of alignment

校准的目的是通过确定雷达天线轴线与车辆行驶轴线在水平及垂直方向上的偏差，再通过算法补偿使二者重合或在一个允许的范围内。

The alignment has the purpose to determine the horizontal and vertical deviation angle of the radar antenna axis to the alignment reflector and thus to the driving axis of the vehicle, and then compensate the deviation angle by a certain algorithm.

5.2 校准工位 Alignment station

一般的雷达校准平台如图 2 所示，包括被校准车辆、校准设备（反射板、反射锥）以及校准控制电脑等。对设备、环境的具体要求，将在后文详细描述。

Basic alignment station setup can take Figure2 as reference, generally include vehicle, alignment equipment (reflective plate or corner reflector) and alignment computer. The detailed requirements will be illustrated in following chapters.

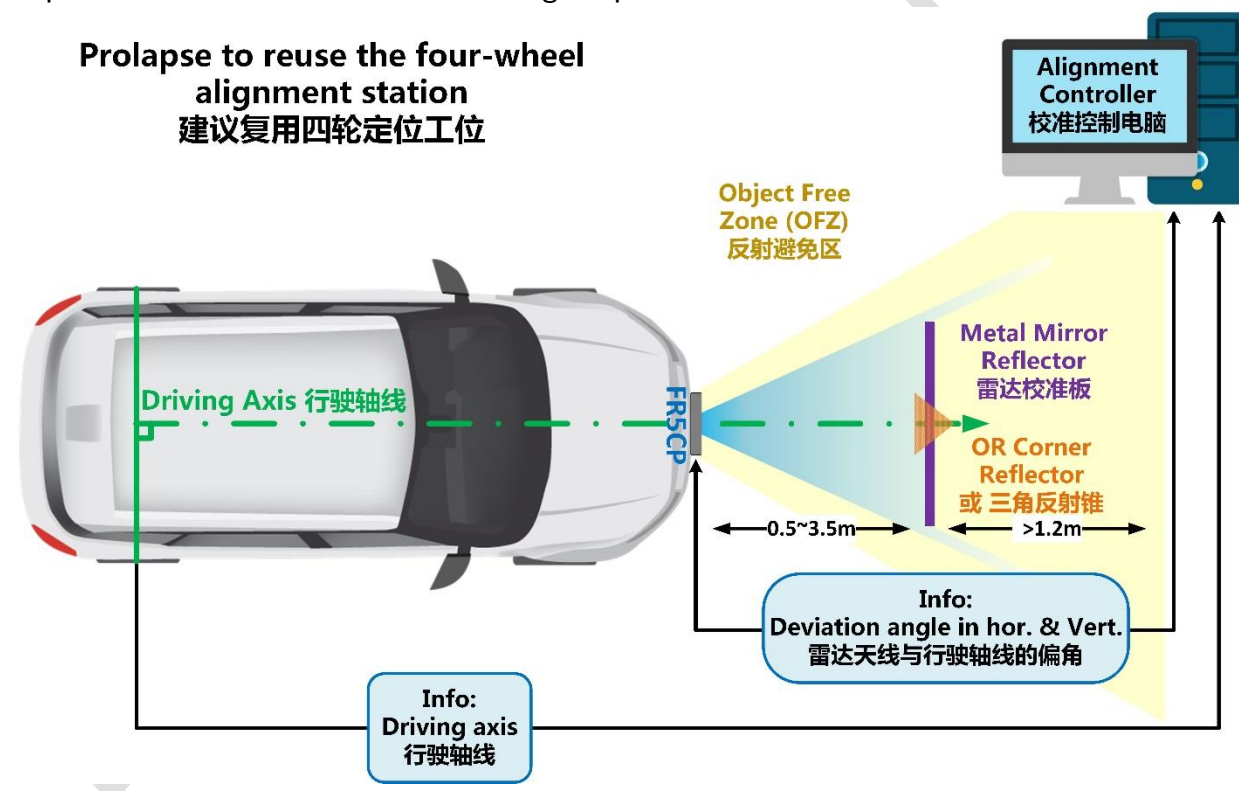


图 2：基本的校准工位搭建

Figure 2: Basic alignment station setup

5.3 静态校准的方式 Static Alignment method

静态校准采用对雷达正前方参考目标的一点式测量来计算雷达天线当前较标准位置的偏置角度。一点式校准所采用的参考目标可以是：

- 平面反射板或反射镜

- 三角反射锥

The static calibration is done using a 1-point measurement with a reference target in front of the sensor. The platform provides the 1-point measurement in combination with the following reference targets:

- Flat plate/ mirror
- Corner reflector

参考目标的角度必须已知，该角度是计算雷达天线偏置角度的基础。下述测量方式都建立在车辆处于规定的反射避免区的基础上。一般来讲，雷达对于自身角度的偏置是基于雷达波对参考目标角度的测量值反算得到的。测量时间与雷达工作在标准模式下的刷新周期一致，在实际工位的节拍设计上，还需考虑由于通讯诊断所产生的延迟。

The expected angle of the reference target is known and is used to determine the misalignment of the sensor. Below the measurement conditions are defined by the reflection free area.

The calculation of the misalignment is based on the angle estimation of the radar sensor. Therefore, the measured vertical and horizontal angle of the reference target are used.

The measurement time is the same as the cycle time in the normal mode. Additionally, there are delays and latencies due to communication and diagnosis.



图 3：采用反射板或反射锥的一点式校准

Figure 3: Measurement set-up 1-point alignment with a corner reflector or flat plate/mirror

一点式静态校准工位搭建的关键要点一般有：

- 雷达天线罩表面距反射板或反射锥在车辆行驶轴线方向的距离；
- 反射避免区的设置；
- 雷达中心点据地面的垂直距离；
- 反射板或反射锥的属性。

Important parameters of the measurement set-up are the following:

- Distance between reference target and sensor;
- Reflection free area;
- Mounting height of the sensor above ground;
- Properties of the reference target.

5.4 校准的流程 Procedure of alignment

静态校准一般流程如图 3 所示，雷达通过分析从特定距离范围内中心高度与雷达安装高度等高的三角锥或反射板收到的回波，可计算得出雷达在水平和垂直方向上与车辆行驶轴线之间的偏差角（即与反射板轴线的偏差）。

进行校准时，系统会同时对测量结果进行可信性检查（如反射板不存在或安装不正确等），如果发生错误，将向诊断仪或校准控制电脑发送错误信息。

如果测量结果可信，则可通过该结果计算出雷达水平和垂直方向的安装偏差角，该偏差角由诊断仪或校准电脑写入雷达 EEPROM 后，雷达将据此进行算法补偿，以达到对目标的准确探测。

The standard procedure of static alignment is shown in Figure 3. The measurement of the horizontal and vertical angle deviation to the alignment reflector (and thus to the vehicle axis) are taken at predefined mirror position.

A plausibility check of the measured values is able to identify the appearance of mistakes during the execution (e.g. reflector not existing or not mounted correctly). If that is the case, an error message is given to the diagnosis tester.

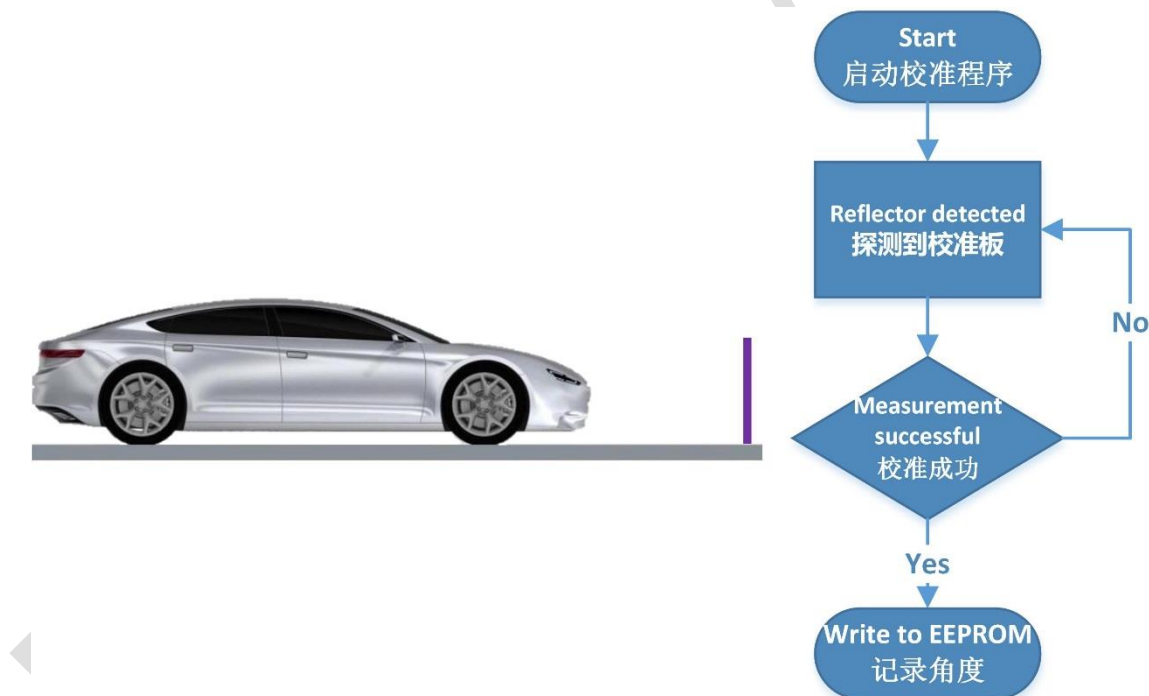


图 4：校准流程示意图

Figure 4: Schematic of alignment procedure

If the measurements are plausible, these are evaluated and the horizontal and vertical angle deviation are calculated. On the basis of the measured angles, radar sensor is able to compensate the measurement deviation in order to guarantee the performance of its functions.

通过诊断故障码“雷达未校准”是否被自动清除来判断校准是否成功。如果不成功，需要重新校准、检查结果。如果多次校准后仍不成功(至少 2 次，具体校准次数取决于生产节拍)可定义为故障，应将车辆送至返修区，用水平仪等设备检查安装情况并进行相应方式调整。

Whether the alignment successful or not, could be checked by the diagnostic trouble code 'Alignment_Not_OK', the alignment is only succeed when DTC automatically cleared. If not, please check the environment and launch the alignment process again.

If the DTC still exists after several rounds of alignment (At least 2 rounds, depends on actual production takt), the very vehicle shall be sent to repair area, and check or adjust the mounting situation by gradienter or other tools.

完整的校准参考时长应在产线校准工位搭建完成后进行测量，单次校准时间以实测为准。主机厂可向校准设备供应商咨询该工位单次校准所需总时长的参考数据。

The actual alignment cycle time only could be measured after EOL station build up, as different environment will leads to different cycle time. OEM should consult the equipment supplier to get the reference cycle time.

5.5 校准的要求 Requirement of alignment

5.5.1 校准场地 Alignment area

校准场地要求如校准工位的坡度，光照条件，地面的平整度等，以设备供应商提出的要求为准，满足博世雷达的校准要求即可。

For the requirement of the alignment area such as the alignment area slope, light condition, unevenness of the floor etc. should follow equipment supplier's suggestion to meet the alignment requirement of BOSCH radar.

5.5.2 参考目标的要求 Reference target specification

对于反射板、反射镜的一般要求参见表 1:

The most important properties of the flat plate/mirror are summarized in table1:

反射板、反射镜位置	反射板、反射镜的法线应与行驶轴线平行，误差 $\pm 0.15^\circ$ ，几何中心需与雷达中心位置一致，误差 $\pm 50\text{mm}$
General position	The geometric center of the plate should be the same as radar center, tolerance $\pm 50\text{mm}$, normal should parallel to driving axis, tolerance $\pm 0.15^\circ$
反射板、反射镜的宽高	35cm x 35 cm 至 70 cm x 70 cm
Height and width	35cm x 35 cm to 70 cm x 70 cm
表面均匀度	< 0.4mm (波长的 10%)
Evenness	< 0.4mm (10% of wave-length)
反射板、反射镜材料	有金属涂层的玻璃或金属

Material	Metalized glass plate or metal
金属层厚度	>100μm
Thickness of metal	>100μm
反射板、反射镜边缘及底座	反射板、反射镜边缘及底座的回波衰减应大于 40dB
Plate edges & reflector stand	Reflections from edges and reflector stand shall have an attenuation >40dB

表 1：反射板、反射镜的要求
Table 1: EOL Flat plate/ mirror specification

对于三角反射锥来讲，主要由雷达散射截面积 RCS 值来评价。三角反射锥的选型取决于雷达天线罩到三角反射锥在车辆行驶轴线方向上的距离，对于不同距离下的不同 RCS 要求请依据表 2 中的总结。

The corner reflector is mainly specified by its RCS value. It is recommended to choose the type of the reference target in dependence of the distance between sensor and reference target, which is summarized in table2.

距离 Distance	0.5m-1m	1m-1.8m	1.8m-2.5m	2.5m-3.5m
一般环境的目标	反射板、反射镜	反射板、反射镜	三角反射锥	三角反射锥
Target in normal environment	Flat plate/mirror 50cm x 50 cm	Flat plate/mirror 70cm x 70 cm	Corner Reflector RCS=3 m²	Corner Reflector RCS=10 m²

表 2：推荐的校准参考目标
Table 2: Reference target recommendation

备注：淡绿色标记的 0.5 至 1m 距离及其所采用的以反射板、反射锥为参考目标的组合是博世推荐的校准工位搭建方式，该方式有着最高的精确度及对于环境干扰的鲁棒性，如地面反射等。

对于反射板、反射镜，其法线应与车辆行驶轴线平行，任何偏差都会导致雷达对自身角度的估算产生额外误差。对于三角反射锥，其锥顶应与雷达中心等高，其锥顶与雷达中心连线应与车辆行驶轴线平行，任何偏差都会导致雷达对自身角度的估算产生额外误差。

误差角可由下述公式计算得出：

- 误差角 = $\arctan(\text{高度误差 [mm]} / \text{雷达天线罩距反射物水平距离 [mm]})$

Remark: The marked range (0.5m – 1m) together with flat plate/mirror is recommended by Bosch due to the highest accuracy and the highest robustness against environmental disturbances (e.g. ground reflections). For flat plate/mirror, the normal of target surface should parallel to the driving axis. Deviations from the expected relative orientation lead to additional measurement errors of the misalignment.

For corner reflector, positioning of the center of the target should orientated with respect to the center of the antenna diagram. Vertical and/or horizontal deviations of the expected position lead to additional measurement errors of the misalignment. They can be calculated:

- Angle_Error [°] = arctan (Error_height [mm] / distance_radar_target [mm])



如须选用三角反射锥，其锥顶应与雷达中心等高。
 If a corner reflector is been choosen instead of a mirror, the middle of the radar antenna has to have the same height like the middle of corner reflector.

图 5: FR5CP 的中心点

Figure 5: Center of FR5CP

5.5.3 环境条件 *Ambient conditions*

FR5CP 的校准需要在场地布置上考虑反射避免区，以规避不期望的回波反射，如金属柱、楼梯等，具体来讲在水平 $\pm 65^\circ$ 及竖直 $\pm 15^\circ$ 范围内，应保证纯粹的反射环境。

The FR5CP alignment concept requires an object free zone (OFZ) to make sure that no unwanted reflections (e.g. from metal posts, floor steps) influence the horizontal and vertical angle estimation inside $\pm 65^\circ$ horizontal and $\pm 15^\circ$ vertical.

反射避免区 Object Free Zone	径向 Radial	在雷达与反射板之间(0.5~3.5 米) 以及反射板后方 1.2 米的区域内，尤其应注意此区域内不允许有金属物体，如金属地面,金属支架等； No obstructions between the sensor and the reflector as well as 1.2m behind the reflector. Especially no obstructions or metal in this area, such as metal ground or metal shelf etc.
	垂直方向 Vertical	$\pm 15^\circ$
	水平方向 Horizontal	$\pm 65^\circ$ 或 1.5x 车宽 $\pm 65^\circ$ or 1.5 x vehicle width

表 3: 反射避免区的要求

Table 3: Requirement of Object Free Zone

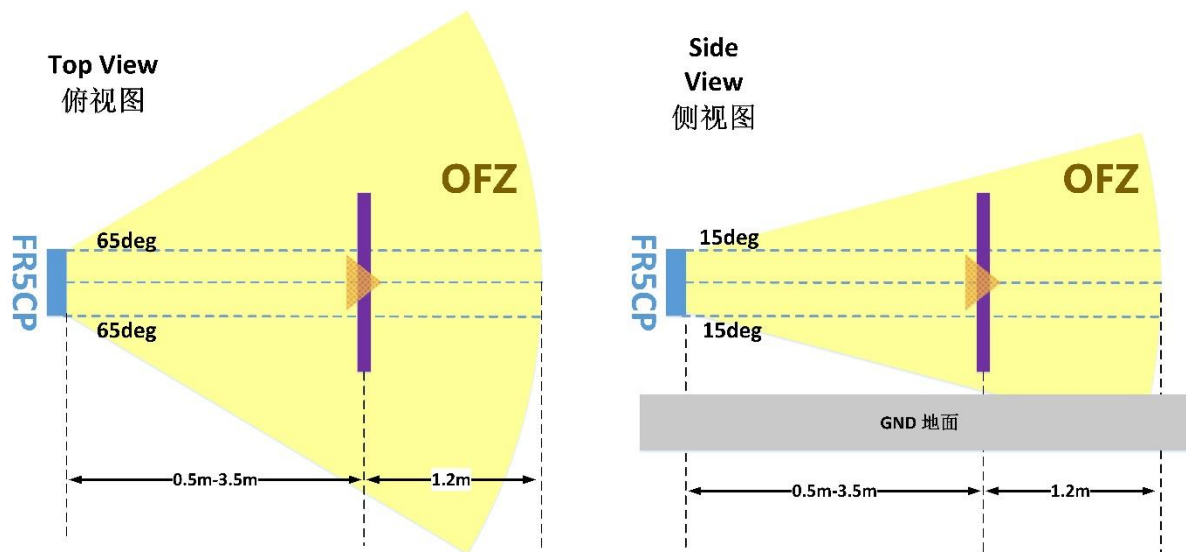


图 6: 反射避免区图示

Figure 6 Object Free Zone

备注：该反射避免区仅为建议，具体与雷达回波的扰动有关。

Remark: The object free zone is only a recommendation. Relevant is the received signal-to-disturbance relation.

对于有无法规避的强干扰的环境，建议采用反射板、反射镜作为参考目标，这种方式的一大优点在于其可以遮盖反射板、反射镜后方的干扰源，同时用吸波材料对干扰物进行覆盖。在这种情况下，反射板、反射镜的宽高尺寸应满足：

最小宽度 = $\tan(2 \times (15^\circ + \beta)) \times \text{雷达天线罩表面与反射板在行驶轴线上的距离} [\text{m}]$

最小高度 = $\tan(2 \times (15^\circ + \varphi)) \times \text{雷达天线罩表面与反射板在行驶轴线上的距离} [\text{m}]$

Min horizontal size [m] = $\tan(2 \times (15^\circ + \beta)) \times \text{distance_radar_target} [\text{m}]$

Min vertical size [m] = $\tan(2 \times (15^\circ + \varphi)) \times \text{distance_radar_target} [\text{m}]$

β = 产线中可能造成的最大雷达水平角度偏差 max. Horizontal misalignment angle of sensor in the plant

φ = 产线中可能造成的最大雷达垂直角度偏差 max. Vertical misalignment angle of sensor in the plant

5.5.4 车辆 Vehicle

雷达校准对车辆有如下要求：

- 确保雷达校准前车辆已做好四轮定位。
- 正确的胎压。
- 车辆稳定的停在转鼓上。
- 校准时，车辆空载。

- 校准完成后车身重量不允许有过大变动，从而造成车辆行驶轴线的偏转；
（例如，在校准后进行车辆进行改造，安装液化器罐等，会造成行驶轴线垂直方向的角度变化）
- 驻车制动激活(可选)。

Radar alignment have requirement to vehicle as below:

- Before radar alignment, vehicle should have finished four wheel alignment.
- Correct tire pressure
- Vehicle should stay on drum stable.
- When alignment, vehicle is unload.
- no heavy parts that might change the driving axis shall be assembled after the alignment
(For example, liquid gas tank)
- Parking brake active (optional)

5.6 客户项目产线校准实际参数 Customer project EOL alignment data（依据 YYYY.MM.DD 最新参数进行计算）

校准板高度 (反射板相对雷达中心) Altitude (Around center)	反射板相对雷达中心±XXX mm Around center ±XXXmm
校准板宽度(反射板相对雷达中心) Breadth (Around center)	反射板相对雷达中心±XXX mm Around center ±XXX mm
反射板平面度 Evenness of the reflector plate/mirror	< 1/10 波长, 0.4 mm <1/10 of wave length, 0.4 mm
反射板材料 Plate material	金属玻璃 metalized glass plate
金属板或金属涂层厚度 Thickness of metal board	>100μm
反射板倾斜角 Reflector misaligned angel	0°
前轴中心点到校准参考目标的距离 Distance from frond axes to reflector	XXXmm(依据 XX 客户提供的最新数据) XXXmm(based on XX customer latest data)
校准参考目标中心水平位置 Reflector center position. Horizontal	雷达中心位置（见图 5） in the center of Radar (see fig.5)
校准参考目标中心垂直位置 Reflector center position. Vertical	雷达中心高度 Radar center height
雷达中心高度 Radar center height	前轴轮眉高度均值+雷达中心距轮眉高度 Average of from axes wheel housing edge height + Radar center to wheel housing edge height

前轮轮眉高度平均值 average wheel housing edge height of front wheels	每台车通过供应商设备实测 Test by supplier equipment for each car
雷达中心距轮眉高度 The height from Radar center to wheel housing edge	需 customer 提供(need to be offered by customer)
参考目标据雷达天线罩表面距离 Distance from radar to reflector	XXX mm(依据 customer 提供的最新数据) XXX mm(based on customer latest data)

表 4: **实际客户项目** 的校准参数

Table 4: **Actual customer project** alignment data

注意:

- 若由于雷达安装高度较低而导致反射板下边沿在地面线以下, 则最少应保证反射板下边沿距离地面 **30mm**, 此时若不能满足反射板几何中心和雷达中心等高的要求, 则需要通过博世进行确认;
- 在标定时, 反射板的法线需与车辆行驶轴线保持平行, 误差 $\pm 0.15^\circ$;
- 如果车辆完成校准后从前方驶出, 主机厂要考虑反射板的移动;
- 反射板边沿应尽可能做倒角处理, 以避免反射板锋利角度造成的反射;
- 以上所有参数均基于 **XX 客户** 提供的基本信息(**YY.MM.DD**)计算得出, 如实际参数与该信息不符, 风险不由博世承担相应参数须重新进行确认。

Attention:

- The target board bottom must higher than 30mm to the ground, If in this case the target board geometric center cannot reach the radar center(due to the radar mounting position too low, the distance between geometric center and radar center need to be confirmed and calculated by BOSCH, meanwhile additional tests needed ;
- The operation method of the reflector (automatic or manual) depends on OEM and equipment supplier.
- Reflector normal and academic driving axis should be in parallel, tolerance $\pm 0.15^\circ$.
- If vehicle move forward after EOL, OME need consider the removing of reflector.
- Chamfer is recommended for the target board in order to avoid sharp corner.
- All parameter of the reflector is based on **XX customer** basic data and key information (**YY.MM.DD**), if the real parameter changed or different, all data need to be updated, any risk is not BOSCH responsibility.

5.7 校准的误差 Inaccuracy of alignment

雷达校准的误差包括测量的误差和校准设备的误差等

The resulting alignment error include Inaccuracy of the measurement and inaccuracy of alignment equipment.

5.7.1 测量的误差 *Inaccuracy of the measurement*

测量的误差包括如下因素：

The error measurement include:

误差组成 Error component	水平方向 Horizontal	垂直方向 Vertical	责任方 Responsible
FR5CP installation tolerance	$\pm 3^\circ$	$\pm 3^\circ$	OEM
雷达安装误差			
主动式校准的不准确度（不含温度及寿命影响） inaccuracy of active alignment measurement (w/o influence due to temperature, lifetime)	$\pm 0.5^\circ$	$\pm 0.5^\circ$	博世 Bosch
反射板法线与行驶轴线最大偏差 Maximum deviation of mirror normal and driving axis	$\pm 0.15^\circ$	$\pm 0.15^\circ$	OEM
反射板垂直角度误差 Inaccuracy of reflect plate vertical angles	n/a	0.05°	OEM

表 5：校准的误差

Table 5: Measurement Error:

注意 Note:

- 当雷达安装于保险杠、覆盖件、隔栅后方及有其他目标反射干扰时，可能还会产生额外的误差。

Additional inaccuracies may arise due to installation behind bumper, cover, grill, reflection from other objects.

- 雷达安装误差 ($\pm 3^\circ$) 是指将雷达安装在支架上，在校准之前，雷达天线轴线与车辆行驶轴线的夹角

Radar installation tolerance is the deviation tolerance which between radar and bracket installation reference surface, before alignment start.

- 请注意反射板的法线与车辆行驶轴线（由非转向轮的前束角决定，和转向轮无关，见图 7）的偏差，博世要求行驶轴线与反射板法线的偏差值可测，且在雷达校准之前该值必须保证在 $\pm 0.15^\circ$ 范围内。需要注意的是，如果车辆在调整自身参数（如调整前束）时改变了行驶轴线，则需要在校准开始之前重新进行转鼓，保证行驶轴线与反射板法线的偏差在 $\pm 0.15^\circ$ 范围内。

Please make sure the Maximum deviation of mirror normal and driving axis (driving axis determined by the toe angle of non-steering wheel, see fig.5) could be measured by the alignment station, and the threshold should be within $\pm 0.15^\circ$. If driving axis has been changed during toe angle alignment or other parameter changing, the car should be revolved again on drum tester to make sure the Maximum deviation of mirror normal and driving axis within $\pm 0.15^\circ$.

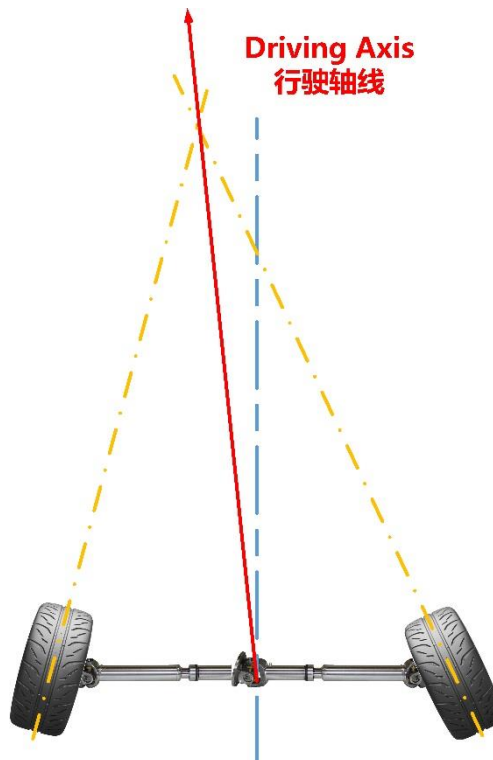


图 7: 行驶轴线的定义

Figure 7 Definition of driving axis

5.8 校准的失效原因 Known limitations

如果有以下情况发生，主动式校准可能会失败

The active alignment measurement may fail or may work incorrectly if

- 反射板没有置于雷达前方 alignment reflector is not positioned in front of the sensor
- 向反射板发射的雷达波被遮挡 the radar beam to the alignment reflector is blocked
- 雷达的安装偏差过大 If the sensor is excessively strongly misaligned
- 有来自其他与反射板径向距离类似的目标的反射干扰
Reflections from other objects with similar radial distance as reflector plate
- 在目标区域内有来自其他目标的更强烈的反射
Stronger reflection from another target inside search area
- 雷达天线特性由于反射板和雷达间其他物体的干扰发生了改变
The antenna diagram is changed by something between sensor and reflector plate

6 诊断功能 Diagnostic Functions

中距离雷达在交付客户时将附带有以下的默认配置

The FR5CP is delivered to the customer with the following coding:

- 产线模式 production mode(defined by SW-PCM)
- 雷达未校准 Alignment not OK(defined by SW-PCM)

产线模式将自适应巡航控制及预测性紧急制动系统功能关闭以确保产线上不会有误触发，该模式同时将雷达波调制设置为适合校准的方式，雷达被设置为未校准状态以防止交付未校准的雷达。

The production mode disables all ACC and PEBS functions to make sure that no error activations occur on the assembly line. It also sets the radar modulation suitable for alignment. The ECU is coded as misaligned as a precaution against delivering an ECU without EOL alignment.

6.1 校准流程 Detailed process and describe

6.1.1 产线校准流程图 General procedure of FR5CP Static Calibration (flow chart modified by SW-PCM)

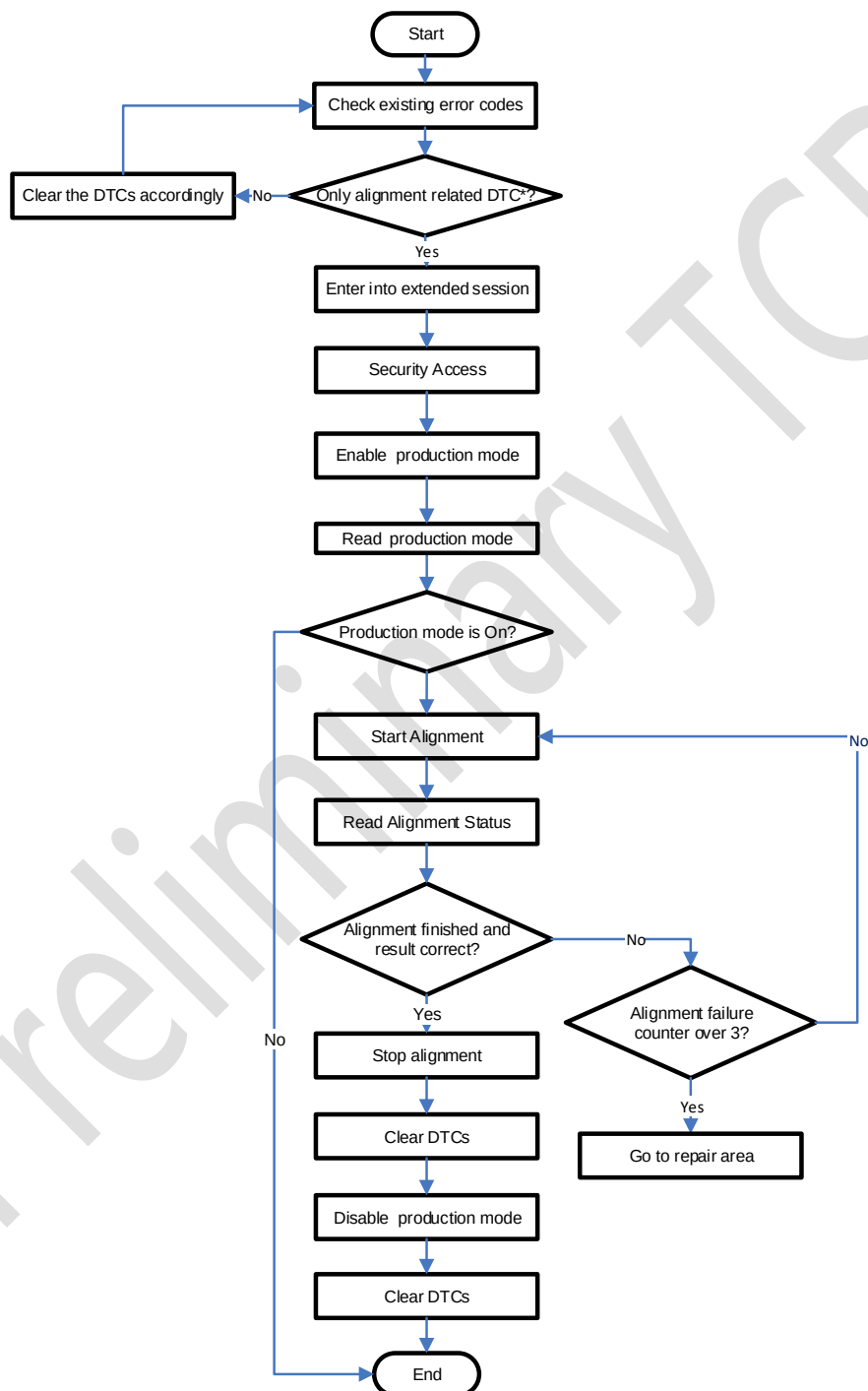


图 8：FR5CP 的校准流程

*: DTC_PRODUCTION_MODE_ACTIVE, DTC_ALIGNMENT_NOT_OK, DTC_MRR_MISALIGNED, DTC_ALIGNMENT_NEVER_DONE

Figure 8: Process of FR5CP calibration

该流程图仅为推荐流程；实际流程的制定需要依据主机厂要求，从而保证工位的正常工作；

this work flow is just recommended work flow ; work flow should be defined by OEM to make sure the station work well.

6.1.2 产线校准步骤 General procedure of FR5CP Static calibration

序号 No.	设备操作 Manual operation	数据流（十六进制） Data flow (HEX)	说明 Description
01	->Ignition Off ->Connect diagnostic Interface ->Ignition On	781 02 10 03 CC CC CC CC CC 789 06 50 03 00 32 00 C8 CC	进入扩展模式 Enter extended mode
02	/	781 02 27 01 CC CC CC CC CC 789 06 67 01 Seed#0 #1 #2 #3 CC 781 06 27 02 Key#0 #1 #2 #3 CC 789 02 67 02 CC CC CC CC CC	安全访问 算法请参考[诊断规范的KEY-SEED 算法] Security access Reference to KEY-SEED算法
03	/	需要定义客户所需信息 TBD by customer XX	读ECU 的ID Read the ID of ECU
04	/	781 04 2E 11 C0 01 CC CC CC 789 03 6E 11 C0 CC CC CC CC	打开ECU的Production Mode，设 为01 Set production mode of ECU 01
05	/	781 03 22 11 C0 CC CC CC CC 789 04 62 11 C0 XX CC CC CC	读ECU的Production Mode,如果xx 为1则继续下面步骤，否则退出测 试 Read the production mode of ECU, if xx is 1, then continue the following steps, or abort the test

06	探测参考目标 Measure the reflector	781 04 31 01 11 D3 CC CC CC 789 04 71 01 11 D3 CC CC CC	开始校准 Start alignment
07	读取并写入偏置角度 Read & Write misaligned angle	781 04 31 03 11 D3 CC CC CC 789 00 09 71 03 11 D3 X1 X2 X3 X4 X5 CC	读取并写入角度 Read & Write X1-X5 各字节含义如下: The meaning for each byte as follows: X1: Bit0-Bit3, Routine Status; 0x00: Routine inactive; 0x01: Routine active; 0x02: Routine NVM write not OK; 0x03: Routine timeout; 0x04: Routine finished correctly; 0x05: Routine aborted; Bit4-Bit7: Routine Result; 0x00: Alignment no result available; 0x01: Alignment incorrect result; 0x02: Alignment correct result; X2 X3 表示 Horizontal deviation angle; X4 X5 表示 Vertical deviation angle. Horizontal deviation angle = X2X3 * 0.01; Vertical deviation angle = X4X5 * 0.01.
08	停止 Stop	781 04 31 02 11 D3 CC CC CC 789 04 71 02 11 D3 CC CC CC	停止校准 Stop alignment

Add Finish and Repeat button to confirm adjust operation.
 在操作界面上增加“完成校准”，“重复校准”按键。用于确认现场操作。

09	Finish 按钮		a.如果步骤07或08的返回值不为00 00的情况下，如果按finish按钮,则在界面进行提示:校准未完成，请重新校准，并等待repeat按钮响应(再按finish按钮无响应); if step 07 or 08 return value xx or yy is not zero, when press finish button, warning : alignment not done, please repeat alignment, then waiting repeat button response(no response with finish button); b.如果步骤07和08返回值均为00 00，跳转至步骤11; if step 07 and 08 both return value xx and yy is zero, when press finish button, jump to step 12
10	Repeat按钮		按repeat按钮后，跳转至步骤6 when press repeat button, jump to step 6
11	/	781 04 14 FF FF FF CC CC CC 789 01 54 CC CC CC CC CC CC	清除DTC Clear DTC
12	/	781 04 2E 11 C0 00 CC CC CC 789 03 6E 11 C0 CC CC CC CC	关闭ECU的Production Mode，设为00 Set production mode of ECU 00
13	/	781 04 14 FF FF FF CC CC CC 789 01 54 CC CC CC CC CC CC	清除DTC Clear DTC

表 6：FR5CP 静态校准的一般步骤
Table 6: General procedure of FR5CP static calibration
注意 Warning :

- 如果雷达校准后还有转鼓测试，“**Production Mode 设置为 0**”，需要在转鼓测试后进行，以避免雷达功能误触发。
 If there are some other tests on the drum after FR5CP alignment, the **“Production Mode set to 0”** should be done only after drum test, in case the misact of radar function.

2. 打开和关闭工厂模式以及写入 VIN 码的命令可依据主机厂要求进行修改；打开关闭工厂模式以及写入 VIN 的命令需要在扩展模式以及安全访问后进行：

Enable/disable production mode write VIN code step could be adjusted by OEM, but should be done only after enter extended mode and security access:

->Ignition Off ->Connect diagnostic Interface ->Ignition On	781 02 10 03 CC CC CC CC CC 789 06 50 03 00 32 00 C8 CC	进入扩展模式 Enter extended mode
/	781 02 27 01 CC CC CC CC CC 789 06 67 01 Seed#0 #1 #2 #3 CC 781 06 27 02 Key#0 #1 #2 #3 CC 789 02 67 02 CC CC CC CC CC	安全访问 算法请参考[诊断规范的KEY-SEED算法] Security access Reference to KEY-SEED算法

在进入扩展模式以及安全访问后可进行工厂模式的更改以及 VIN 的写入：

After enter extended mode and security access, production mode could be changed and VIN is possible to write in the sensor:

打开工厂模式：enable production mode(function disabled)

/	781 04 2E 11 C0 01 CC CC CC 789 03 6E 11 C0 CC CC CC CC	打开ECU的Production Mode, 设为01 Set production mode of ECU 01
---	--	--

关闭工厂模式：disable production mode(function enabled)

/	81 04 2E 11 C0 00 CC CC CC 789 03 6E 11 C0 CC CC CC CC	关闭ECU的Production Mode, (production mode设为00) Set production mode of ECU 00
---	---	---

写入 VIN 码：write VIN code

/	781 00 14 2E F1 90 XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX CC CC	写入VIN码 Write VIN Code
---	---	--

6.2 诊断测试 Diagnostic Test

第一步的目的是确保雷达在安装至车辆后，在校准前工作正常。

The reason for this first step is to ensure that the FR5CP is working correctly after vehicle assembly before moving on to alignment.

6.2.1 进入扩展诊断会话 **Start Extended Diagnostic Session**

进入扩展诊断会话 Start Extended Diagnostic Session	
操作员 Operator	诊断仪/雷达控制器 Tester /Radar ECU
⇒ 熄火 Ignition Off ⇒ 连接诊断接口 Connect diagnostic Interface ⇒ 点火 Ignition On ⇒ 等待直至雷达系统测试完成 Wait until FR5CP system test is finished	诊断仪 Tester: 进入扩展诊断会话 Start extended diagnostic session

诊断仪和雷达之间的通信是基于客户定义的诊断规范完成。
 The communication set up between the Tester and ECU is based on the customer diagnostics according to the corresponding specification.

通信失败 Communication not successful ⇒ 设置失败信息，中止测试 set failure message, abort test

通信成功 Communication successful ⇒ 继续 continue

6.2.2 读 ECU 的 ID **Read ECU-ID**

读 ECU 的 ID Read ECU-ID	
操作员 Operator	诊断仪/雷达控制器 Tester /Radar ECU
⇒ 读 ECU 标识符 Read ECU Identification	诊断仪 Tester: 请求 ECU 识别码 Request ECU Identification 雷达控制器 ECU: 输出 ECU 识别码 Output ECU Identification + customer required information

ECU标识符的目的是为了检查雷达是否与车辆相匹配，以及避免操作员将不合适的雷达装配上去。ECU标识符须经过客户及其装配线确认，该标识符须与下线检测设备软件中存储的ECU数据相比较。

为保证产品的可追溯性，需要将车辆VIN码和FR5CP的序列号对应起来。

The intention of ECU identification is checking whether FR5CP can match the vehicle or not, and preventing operator from assembling unsuitable FR5CP. The ECU identification data to be evaluated must be confirmed by the customer and its assembly plants. The ECU identification

data must be compared with the in the end of line test equipment software stored desired ECU data.

To make sure every ECU can be carried up. The FR5CP series Number should be connected with VIN.

不可知的系统 Unknown System ⇒ 中止测试 Abort Test

检测到系统 System detected ⇒ 继续 Continue

Customer project FR5CP 项目:

读取序列号, 并和 VIN 绑定, 在校准之前的“最终线”进行。

Customer project FR5CP Project:

Read FR5CP series Number and combine VIN at the position “Final Line” which happen before FR5CP calibration.

6.2.3 产线模式 *Production Mode*

读取产线模式 Read Production Mode

操作员 **Operator**

诊断仪/雷达控制器 Tester /Radar ECU

⇒ 确认 ECU 是处于产线模式
Confirm that ECU is in production mode

诊断仪 Tester:
 请求 ECU 的模式 Request ECU mode
 雷达控制器 ECU:
 输出 ECU 的模式 Output ECU mode

其它产线模式 Other production mode ⇒ 中止测试 Abort Test

ECU 处于产线模式 ECU in production mode ⇒ 继续 Continue

6.2.4 清除/读取诊断故障码 *Clear/Read DTCs*

清除/读取诊断故障码 Clear/Read DTCs

操作员 **Operator**

诊断仪/雷达控制器 Tester /Radar ECU

⇒ 清除及读取诊断故障码 **Clear and Read DTCs**

诊断仪 Tester:
 清除诊断故障码 Clear DTCs
 请求诊断故障码 Request DTCs
 雷达控制器 ECU:
 删除诊断故障码 (包括历史故障码)
 Delete DTCs (including history)
 输出诊断故障码 Output DTCs

在此步骤仅允许显示以下诊断故障码

At this stage the following DTCs only may be reported:

- 产线模式激活 Production mode active
- 雷达未校准 Sensor Misaligned

- 其它需客户确定，如安装安全气囊等 To be defined by customer depending when e.g. Airbag is attached

有产线模式及雷达未校准之外的其它诊断故障码

DTCs other than Production active and Sensor misaligned ⇒ 中止测试 Abort Test

没有其它剩余的诊断故障码 No other residual DTCs ⇒ 继续 Continue